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5 Vs of Big Data

The 5 Vs of big data (velocity, volume, value, variety and veracity) are the five main and innate characteristics of big data.

12-Factor App Design

A methodology for building modern, scalable, maintainable software-as-a-service applications.

2-Factor or 2-Step Authentication

Two-Factor Authentication, also known as 2FA or TFA or Two-Step Authentication is when a user provides two authentication factors; usually, firstly a password and then a second layer of verification such as a code texted to their device, shared secret, physical token, or biometrics.

A/B Testing

Deploy different versions of an EUT to different customers and let the customer feedback determine which is best.

A3 Problem Solving

A structured problem-solving approach that uses a lean tool called the A3 Problem-Solving Report. The term "A3" represents the paper size historically used for the report (a size roughly equivalent to 11" x 17").

Acceptance test

It assesses whether the entire system meets business requirements and user expectations. This testing ensures that the software is ready for deployment and aligns with customer needs, reducing risks before release.

Access Management

Granting an authenticated identity access to an authorized resource (for example data, service, environment) based on defined criteria (for example a mapped role), while preventing unauthorized identity access to a resource.

Access Provisioning

Access provisioning is the process of coordinating the creation of user accounts, e-mail authorizations in the form of rules and roles, and other tasks such as provisioning of physical resources associated with enabling new users to systems or environments.

Administration Testing

The purpose of the test is to determine if an End User Test (EUT) is able to process administration tasks as expected.

Advice Process

Any person making a decision must seek advice from everyone meaningfully affected by the decision and people with expertise in the matter. Advice received must be taken into consideration, though it does not have to be accepted or followed. The objective of the advice process is not to form a consensus, but to inform the decision-maker so that they can make the best decision possible. Failure to follow the advice process undermines trust and unnecessarily introduces risk to the business.

Agile

A work management method for complex endeavors that divides tasks into small 'sprints' of work with frequent reassessment and adaptation of plans.

Agile (adjective)

Able to move quickly and easily; well coordinated. Able to think and understand quickly; able to solve problems and have new ideas.

Agile Coach

Helps teams master Agile development and DevOps practices; enables productive ways of working and collaboration.

Agile Enterprise

A fast-moving, flexible, and robust company capable of rapid response to unexpected challenges, events, and opportunities.

Agile Manifesto

A formal proclamation of values and principles to guide an iterative and people-centric approach to software development. http://agilemanifesto.org

Agile Portfolio Management

Involves evaluating in-flight projects and proposed future initiatives to shape and govern the ongoing investment in projects and discretionary work. CA's Agile Central and VersionOne are examples

Agile Practice Owner
A role accountable for the overall quality of a service management practice and owner of the Practice Backlog.

Agile Principles
The twelve principles that underpin the Agile Manifesto.

Agile Process
Delivers "just enough" structure and control to enable the organization to achieve its service outcomes in the most expeditious, effective, and efficient way possible. It is easy to understand, easy to follow, and prizes its collaboration and outcomes more than its artifacts.

Agile Process Engineering
An iterative and incremental approach to designing a process with short, iterative designs of potentially shippable process increments or microprocesses.

Agile Process Improvement
Ensures that IT Service Management agility introduced through Agile Process Engineering is continually reviewed and adjusted as part of IT Service Management's commitment to continual improvement.

Agile Service Management
A framework that ensures that ITSM processes reflect Agile values and are designed with "just enough" control and structure in order to effectively and efficiently deliver services that facilitate customer outcomes when and how they are needed.

Agile Service Management Artifacts
Practice Backlog, Sprint Backlog, Increment

Agile Service Management Events
Practice/microprocess Planning, The Sprint, Sprint Planning, Process Standup, Sprint Review, Sprint Retrospective

Agile Service Management Roles
Agile Practice Owner, Agile Service Management Team, Agile Service Manager.

Agile Service Management Team
A team of at least 3 people (including a customer or practitioner) that is accountable for a single microprocess or a complete service management practice.

Agile Service Manager
An Agile Service Management subject matter expert who is the coach and protector of the Agile Service Management Team.

Agile Software Development
Group of software development methods in which requirements and solutions evolve through collaboration between self-organizing, cross-functional teams. Usually applied using the Scrum or Scaled Agile Framework approach.

AIOps
It is the practice of applying Artificial Intelligence (AI) to IT Operations.

Alert Noise Reduction
In AIOps (Artificial Intelligence for IT Operations) refers to the process of minimizing or eliminating irrelevant or redundant alerts generated by IT systems and infrastructure.

Alerting
The process of notifying operators or developers when a system is behaving unexpectedly or has an issue. Alerts can be triggered based on metrics, logs, or other signals.

Amazon Web Services (AWS)
Amazon Web Services (AWS) is a secure cloud services platform, offering compute power, database storage, content delivery, and other functionality to help businesses scale and grow.

Analytics
Test results processed and presented in an organized manner in accordance with analysis methods and criteria.

Anomaly detection
The process of identifying unusual or unexpected behaviour within a system. Anomaly detection can be used to identify issues before they cause significant problems.

Andon

A system gives an assembly line worker the ability, and moreover the empowerment, to stop production when a defect is found, and immediately call for assistance.

Anti-pattern

A commonly reinvented but poor solution to a problem.

Anti-fragility

Antifragility is a property of systems that increases its capability to thrive as a result of stressors, shocks, volatility, noise, mistakes, faults, attacks, or failures.

API Testing

The purpose of the test is to determine if an API for an EUT functions as expected.

Application Performance Management (APM)

APM is the monitoring and management of the performance and availability of software applications. APM strives to detect and diagnose complex application performance problems to maintain an expected level of service.

Application Programming Interface (API)

A set of protocols used to create applications for a specific OS or as an interface between modules or applications.

Application Programming Interface (API) Testing

The purpose of the test is to determine if an API for an EUT functions as expected.

Application Release

Controlled continuous delivery pipeline capabilities including automation (release upon code commit).

Application Release Automation (ARA) or Orchestration (ARO)

Controlled continuous delivery pipeline capabilities including automation (release upon code commit), environment modelling (end-toend pipeline stages, and deploy application binaries, packages, or other artifacts to target environments), and release coordination (project, calendar, and scheduling management, integrate with change control and/or IT service support management).

Application Test-Driven Development (ATDD)

Acceptance Test-Driven Development (ATDD) is a practice in which the whole team collaboratively discusses acceptance criteria with examples, and then distils them into a set of concrete acceptance tests before development begins.

Application Testing

The purpose of the test is to determine if an application is performing according to its requirements and expected behaviours.

Application Under Test (AUT)

The EUT is a software application. For example, business application is being tested.

Architecture

The fundamental underlying design of computer hardware, software, or both in combination.

Artifact

Any element in a software development project including documentation, test plans, images, data files, and executable modules.

Artifact Repository

Store for binaries, reports, and metadata. Example tools include JFrog Artifactory, Sonatype Nexus.

Artificial Intelligence (AI)

It is software that thinks like a human through analysis and reasoning to perform complex tasks.

Attack path

The chain of weaknesses a threat may exploit to achieve the attacker's objective. For example, an attack path may start by compromising a user's credentials, which are then used in a vulnerable system to escalate privileges, which in turn is used to access a protected database of information, which is copied out to an attacker's own server(s).

Audit Management

The use of automated tools to ensure products and services are auditable, including keeping audit logs of build, test and deploy activities, auditing configurations, and users, as well as log files from production operations.

Authentication

The process of verifying an asserted identity. Authentication can be based on what you know (for example password or PIN), what you have (token or one-time code), what you are (biometrics), or contextual information.

Authorization

The process of granting roles to users to have access to resources.

Auto-DevOps

Auto-DevOps brings DevOps best practices to your project by automatically configuring software development lifecycles. It automatically detects, builds, tests, deploys, and monitors applications.

Auto-scaling

The ability to automatically and elastically scale and de-scale infrastructure depending on traffic and capacity variations while maintaining control of costs.

Automatic Instrumentation

A feature in OpenTelemetry that allows developers to easily monitor the performance and behaviour of their applications without having to manually instrument the code with the OpenTelemetry SDKs.

Automated rollback

If a failure is detected during a deployment, an operator (or an automated process) will verify the failure and roll back the failing release to the previous known working state.

Availability

Availability is the proportion of time a system is in a functioning condition and therefore available (to users) to be used.

Backdoor

A backdoor bypasses the usual authentication used to access a system. Its purpose is to grant the cybercriminals future access to the system even if the organization has remediated the vulnerability initially used to attack the system.

Backlog

Requirements for a system expressed as a prioritized list of product backlog items usually in the form of 'User Stories'. The product backlog is prioritized by the Product Owner and should include functional, non-functional, and technical team-generated requirements.

Basic Security Hygiene

A common set of minimum-security practices that must be applied to all environments without exception. Practices include basic network security (firewalls and monitoring), hardening, vulnerability and patch management, logging and monitoring, basic policies and enforcement (may be implemented under a "policies as code" approach), and identity and access management.

Batch Data

Batch processing collects data over time and sends it for processing once collected. It is generally meant for large data quantities that are not time sensitive.

Batch Sizes

Refers to the volume of features involved in a single code release.

Bateson Stakeholder Map

A tool for mapping stakeholder's engagement with the initiative in progress.

Behaviour-Driven Development (BDD)

Test cases are created by simulating an EUT's externally observable inputs, and outputs. Example tool: Cucumber.

Beyond Budgeting
A management model that looks beyond command-and-control towards a more empowered and adaptive state.

Big Data
It refers to data sets that are "too large" or diverse causing traditional data processing techniques to be ineffective.

Black-Box
Test case only uses knowledge of externally observable behaviours of an EUT.

Blameless postmortems
A process through which engineers whose actions have contributed to a service incident can give a detailed account of what they did without fear of punishment or retribution.

Blast Radius
Used for impact analysis of service incidents. When a particular IT service fails, the users, customers, other dependent services are affected.

Blue/Green Testing or Deployments
Taking software from the final stage of testing to live production using two environments labelled Blue and Green. Once the software is working in the green environment, switch the router so that all incoming requests go to the green environment - the blue one is now idle.

Bottleneck
The single point in a workflow where the capacity to process work is less than the demand placed upon it, creating a constraint that restricts the entire value stream's end-to-end flow and determines its maximum throughput.

Bug
An error or defect in software that results in an unexpected or system-degrading condition.

Bureaucratic Culture
Bureaucratic organizations are likely to use standard channels or procedures which may be insufficient in a crisis (Westrum).

Bursting
Public cloud resources are added as needed to temporarily increase the total computing capacity of a private cloud.

Business Case
Justification for a proposed project or undertaking on the basis of its expected commercial benefit.

Business Continuity
Business continuity is an organization's ability to ensure operations and core business functions are not severely impacted by a disaster or unplanned incident that takes critical services offline.

Business Transformation
Changing how the business functions. Making this a reality means changing culture, processes, and technologies in order to better align everyone around delivering on the organization's mission.

Business Value
In management, an informal term that includes all forms of value that determine the health and well-being of the firm in the long run.

Cadence
Flow or rhythm of events.

CALMS Model
Considered the pillars or values of DevOps: Culture, Automation, Lean, Measurement, Sharing (as put forth by John Willis, Damon Edwards, and Jez Humble).

Canary Testing
A canary (also called a canary test) is a push of code changes to a small number of endusers who have not volunteered to test anything. Similar to incremental rollout, it is where a small portion of the user base is updated to a new version first. This subset, the canaries, then serve as the proverbial "canary in the coal mine". If something goes wrong then a release is rolled back and only a small subset of the users are impacted.

Capacity
An estimate of the total amount of engineering time available for a given Sprint.

Capacity Test

The purpose of the test is to determine if the EUT can handle expected loads such as number of users, number of sessions, aggregate bandwidth.

Capture-Replay

Test cases are created by capturing live interactions with the EUT, in a format that can be replayed by a tool. For example Selenium

Cardinality

A mathematical term identifying the number of possible values or the number of unique values within a data dimension.

Carrots

Positive incentives, for encouraging and rewarding desired behaviours.

Cassandra

A highly scalable, distributed NoSQL database designed for handling large amounts of data across many servers.

Causal Observability

When you can see which events caused what other events. It is level 3 of the Observability Maturity Model and requires correlation of telemetry (metrics, events, logs, traces) and topology at every moment in time.

Chain of Goals

A method designed by Roman Pichler of ensuring that goals are linked and shared at all levels through the product development process.

Change

Addition, modification, or removal of anything that could have an effect on IT services. (ITIL® definition)

Change Failure Rate

A measure of the percentage of failed/rolled back changes.

Change Fatigue

A general sense of apathy or passive resignation towards organizational changes by individuals or teams.

Change Lead Time

A measure of the time from a request for a change to the delivery of the change.

Change Leader Development Model

Jim Canterucci's model for five levels of change leader capability.

Change Management

The process that controls all changes throughout their lifecycle. (ITIL definition)

Change Management (Organizational)

An approach to shifting or transitioning individuals, teams and organizations from a current state to a desired future state. Includes the process, tools and techniques to manage the people-side of change to achieve the required business outcome(s).

Change-based Test Selection Method

Tests are selected according to a criterion that matches attributes of tests to attributes of the code that is changed in a build.

Chaos Engineering

The discipline of experimenting on a software system in production in order to build confidence in the system's capability to withstand turbulent and unexpected conditions.

Chaos Mesh

A Kubernetes-native Chaos Engineering platform for testing system resilience under failures.

Chaos Monkey

A tool from Netflix that randomly terminates instances to test system reliability.

Chaos Toolkit

An open-source Chaos Engineering framework for automating experiments on systems.

Chapter Lead

A squad line manager in the Spotify model who is responsible for traditional people management duties is involved in day-to-day work and grows individual and chapter competence.

Chapters

A small family of people having similar skills and who work within the same general competency area within the same tribe. Chapters meet regularly to discuss challenges and areas of expertise in order to promote sharing, skill development, re-use, and problem-solving.

ChatOps

An approach to managing technical and business operations (coined by GitHub) that involves a combination of group chat and integration with DevOps tools. Example tools include Atlassian HipChat/Stride, Microsoft Teams, Slack.

Check-in

The action of submitting a software change into a system version management system.

Child span

In tracing, child span is a sub-operation within a trace that records timing and context under a parent span.

CIA (Confidentiality, Integrity, Availability) Triad

An information security model that can guide an organization's efforts and policies aimed at keeping its data secure. The three letters in "CIA Triad" stand for Confidentiality, Integrity, and Availability.

CI/CD Pipeline Scanning

This pipeline execution process provides an opportunity to observe potential security issues you could have in your images. Most CI/CD tools allow integration of additional security scanners.

CI Regression Test

A subset of regression tests that are run immediately after a software component is built. Same as Smoke Test.

Clear-Box

Same as Glass-Box Testing and White-Box Testing.

Cloud Computing

The practice of using remote servers hosted on the internet to host applications rather than local servers in a private data centre.

Cloud-Native

Native cloud applications (NCA) are designed for cloud computing.

Cloud Native Computing Foundation (CNCF)

A Linux Foundation project that supports many useful open-source projects that was founded to help advance container technology.

Cloud Networking

A subset of network monitoring which offers connectivity to applications and workloads across clouds, cloud services and so on.

Cloudbees

Cloudbees is a commercially supported proprietary automation framework tool that works with and enhances Jenkins by providing enterprise levels support and add-on functionality.

Cluster Cost Optimization

Tools like Kubecost, Replex, Cloudability use monitoring to analyze container clusters and optimize the resource deployment model.

Cluster Monitoring

Tools that let you know the health of your deployment environments running in clusters such as Kubernetes.

Clustering

Natural Grouping of unlabeled data or (a group of computers called nodes or members) work together as a cluster connected through a fast network acting as a single system.

Code Coverage

A measure of white box test coverage by counting code units that are executed by a test. The code unit may be a code statement, a code branch, or control path or data path through a code module.

Code Quality

See also static code analysis, Sonar and Checkmarks are examples of tools that automatically check the seven main dimensions of code quality – comments, architecture, duplication, unit test coverage, complexity, potential defects, language rules.

Code Repository

A repository where developers can commit and collaborate on their code. It also tracks historical versions and potentially identifies conflicting versions of the same code. Also referred to as "repository" or "repo."

Code Review

Software engineers inspect each other's source code to detect coding or code formatting errors.

Cognitive Bias

Cognitive bias is a limitation in objective thinking that is caused by the tendency for the human brain to perceive information through a filter of personal experience and preferences: a systematic pattern of deviation from norm or rationality in judgment.

Collaboration

People jointly working with others towards a common goal.

Collaborative Culture

A culture that applies to everyone which incorporates an expected set of behaviours, language, and accepted ways of working with each other reinforced by leadership.

Collective Body of Knowledge (CBoK)

It is an evolving collection of insights from conferences, books, articles, webinars, videos, podcasts, expert experiences, and AI. It spans all aspects of IT, including culture, automation, infrastructure, security, and organizational structure. Unlike fixed IT frameworks, this is dynamic and shaped by the collective contributions of the global DevOps community.

Common Data Model (CDM)

A standardized, predefined set of data schemas and relationships that acts as a universal language, allowing different applications to share and understand information without custom integration.

Compatibility Test

Test with the purpose to determine if an EUT interoperates with another EUT such as peerto-peer applications or protocols.

Configuration Management

Configuration management (CM) is a systems engineering process for establishing and maintaining consistency of a product's performance, functional, and physical attributes with its requirements, design, and operational information throughout its life.

Conformance Test

The purpose of the test is to determine if an EUT complies with a standard.

Constraint

Limitation or restriction; something that constrains. See also 'bottleneck'.

Container

A way of packaging software into lightweight, stand-alone, executable packages including everything needed to run it (code, runtime, system tools, system libraries, settings) for development, shipment, and deployment.

Container Network Security

Used to prove that any app that can be run on a container cluster with any other app can be confident that there is no unintended use of the other app or any unintended network traffic between them.

Container Registry

Secure and private registry for Container images. Typically allowing for easy upload and download of images from the build tools. Docker Hub, Artifactory, Nexus are examples.

Container Scanning

When building a Container image for your application, tools can run a security scan to ensure it does not have any known vulnerability in the environment where your code is shipped. Blackduck, Synopsis, Synk, Claire, and Klar are examples.

Continual Improvement

One of the ITIL® 4 Official Practice Guides and a stage of the service lifecycle.

Continuous Delivery (CD)

A methodology that focuses on making sure software is always in a releasable state throughout its lifecycle.

Continuous Delivery (CD) Architect
A person who is responsible to guide the implementation and best practices for a continuous delivery pipeline.

Continuous Delivery Pipeline
A continuous delivery pipeline refers to the series of processes that are performed on product changes in stages. A change is injected at the beginning of the pipeline. A change may be new versions of code, data, or images for applications. Each stage processes the artifacts resulting from the prior stage. The last stage results in deployment to production.

Continuous Delivery Pipeline Stage
Each process in a continuous delivery pipeline. These are not standard. Examples are Design: determine implementation changes; Creation: implement an unintegrated version of design changes; Integration: merge.

Continuous Deployment
A set of practices that enable every change that passes automated tests to be automatically deployed to production.

Continuous Flow
Smoothly moving people or products from the first step of a process to the last with minimal (or no) buffers between steps.

Continuous Improvement
Based on Deming's Plan-Do-Check-Act, a model for ensuring ongoing efforts to improve products, processes, and services.

Continuous Integration (CI)
A development practice that requires developers to merge their code into trunk or master ideally at least daily and perform tests (that is, unit, integration, and acceptance) at every code commit.

Continuous Integration Tools
Tools that provide an immediate feedback loop by regularly merging, building, and testing code. Example tools include Atlassian Bamboo, Jenkins, Microsoft VSTS/Azure DevOps, TeamCity.

Continuous Monitoring (CM)
This is a class of terms relevant to logging, notifications, alerts, displays, and analysis of test results information.

Continuous Testing (CT)
This is a class of terms relevant to the testing and verification of an EUT in a DevOps environment.

Conversation Café
Conversation Cafés are open, hosted conversations in cafés as well as conferences and classrooms—anywhere people gather to make sense of our world.

Conway's Law
Organizations that design systems are constrained to produce designs that are copies of the communication structures of these organizations.

Cooperation versus Competition
The key cultural value shift toward being highly collaborative and cooperative, and away from internal competitiveness and divisiveness.

COTS
Commercial-off-the-shelf solution.

Critical Success Factor (CSF)
Something that must happen for an IT service, process, plan, project or other activity to succeed.

Cybersecurity Operations Centre (CSOC)
A centralized team monitoring and responding to security threats.

Cultural debt
This is when cultural values, collaboration, or employee well-being are deprioritized in pursuit of rapid growth and innovation, potentially leading to long-term organizational challenges.

Cultural Iceberg
A metaphor that visualizes the difference between observable (above the water) and non-observable (below the waterline) elements of culture.

Culture (Organizational Culture)

The values and behaviours that contribute to the unique psychosocial environment of an organization.

Cumulative Flow Diagram

A cumulative flow diagram is a tool used in agile software development and lean product development. It is an area graph that depicts the quantity of work in a given state, showing arrivals, time in queue, quantity in a queue, and departure.

Current State Map

A form of value stream map that helps you identify how the current process works and where the disconnects are.

Customer Reliability Engineer (CRE)

CRE is what you get when you take the principles and lessons of SRE and apply them to customers.

Cycle Time

A measure of the time from the start of work to ready for delivery. When used in the context of Flow metrics, Cycle Time shows how long individual work items spend in specific workflow states. This is to understand how work flows through individual states (for example, In Progress) or combinations of states (for example, from Ready to Deployed).

Daily Scrum

Daily timeboxed event of 15 minutes or less for the Team to replan the next day of work during a Sprint.

Dashboard

Graphical display of summarized data, for example, deployment frequency, velocity, test results. A visual representation of key metrics and data related to a system. Dashboards can be used to monitor the health and performance of a system in real-time.

DAST (Dynamic Application Security Testing)

Dynamic application security testing (DAST) is a process of testing an application or software product in an operating state.

Data bias

In AI, data bias is the systematic distortion in training data that causes models to produce unfair, inaccurate, or unrepresentative outcomes.

Data fabric

An architecture that provides seamless, unified access to data across distributed environments.

Data Governance

The specification of decision rights and an accountability framework to ensure the appropriate behaviour in the valuation, creation, consumption and control of data and analytics.

Data Loss Protection (DLP)

Tools that prevent files and content from being removed from within a service environment or organization.

DataOps

Defined as a combination of data and engineering practices. It aims in making the data useful by bringing in governance, ingestion guidelines, processing, and consuming structures.

Data Paradox

This means the more data you have, the less information you have.

Data pipeline

A data pipeline refers to a series of processes and workflows that Extract, Transform, and Load (ETL) data from various sources and deliver it to a destination for storage, analysis, or consumption.

Database Reliability Engineer (DBRE)

A person responsible for keeping database systems that support all user-facing services in production running smoothly.

Debugging

The process of finding and fixing problems in a system. Observability tools can be used to help debug issues by providing insights into the system's behaviour.

Defect Density

The number of faults found in a unit. For example, number of defects per KLOC, number of defects per change.

Definition of Done (DoD)

A shared understanding of expectations that an Increment or backlog item must live up to.

DEI (Distributed, Immutable, Ephemeral) Triad

A new paradigm of security and stands for distributed, immutable and ephemeral.

Delivery Cadence

The frequency of deliveries. For example, number of deliveries per day, per week, and so on.

Delivery Package

Set of release items (files, images, and so on) that are packaged for deployment.

Deming Cycle

A four-stage cycle for process management, attributed to W. Edwards Deming. Also called Plan-Do-Check-Act (PDCA).

Dependency Firewall

Many projects depend on packages that may come from unknown or unverified providers, introducing potential security vulnerabilities. There are tools to scan dependencies but that is after they are downloaded. These tools prevent those vulnerabilities from being downloaded to begin with.

Dependency Mapping

It is a tool for analysing dependencies and other relationships between issues in your Jira projects.

Dependency Proxy

For many organizations, it is desirable to have a local proxy for frequently used upstream images/packages. In the case of CI/CD, the proxy is responsible for receiving a request and returning the upstream image from a registry, acting as a pull-through cache.

Dependency Relations

The type of relation on an escalation graph. It describes how components, broadly speaking, affect each other.

Dependency Scanning

Used to automatically find security vulnerabilities in your dependencies while you are developing and testing your applications. Synopsys, Gemnasium, Retire.js, and bundleraudit are popular tools in this area.

Deployment

The installation of a specified version of software to a given environment (for example, promoting a new build into production).

Deployment frequency

It is the rate at which an organization releases code changes to production.

Deployment pipeline

It is an automated process that manages all changes from check-in to production release. It ensures that software progresses through stages such as build, test, integration, and deployment using a toolchain that spans different teams and silos. By automating this process, organizations achieve faster, more reliable, and consistent software delivery.

Design for Testability

An EUT is designed with features that enable it to be tested.

Design Principles

Principles for designing, organizing, and managing a DevOps delivery operating model.

Deterministic Systems

Systems in which the outcome or behaviour can be precisely determined or predicted based on the initial conditions and the rules or laws governing the system. In a deterministic system, there is no randomness or uncertainty involved in the evolution of the system over time.

Dev

Individuals involved in software development activities such as application and software engineers.

Developer (Dev)

An individual who has the responsibility to develop changes for an EUT. Alternate: Individuals involved in software development activities such as application and software engineers.

Development Test

Ensuring that the developer's test environment is a good representation of the production test environment.

Device Under Test (DUT)

The DUT is a device (for example, router or switch) being tested.

DevOps

"A cultural and professional movement that stresses communication, collaboration, and integration between software developers and IT operations professionals while automating the process of software delivery and infrastructure changes. It aims at establishing a culture and environment where building, testing, and releasing software, can happen rapidly, frequently, and more reliably" (Wikipedia).

DevOps Coach

Helps teams master Agile development and DevOps practices; enables productive ways of working and collaboration.

DevOps Infrastructure

The entire set of tools and facilities that make up the DevOps system. Includes CI, CT, CM, and CD tools.

DevOps Kaizen

Kaizen is a Japanese word that closely translates to "change for better," the idea of continuous improvement, large or small, involving all employees and crossing organizational boundaries. Damon Edwards' DevOps Kaizen shows how making small, incremental improvements (little J's) has an improved impact on productivity long term.

DevOps Pipeline

The entire set of interconnected processes that make up a DevOps Infrastructure.

DevOps Score

A metric showing DevOps adoption across an organization and the corresponding impact on delivery velocity.

DevOps Toolchain

The tools needed to support a DevOps continuous development and delivery cycle from idea to value realization.

DevSecOps

A mindset that "everyone is responsible for security" with the goal of safely distributing security decisions at speed and scale to those who hold the highest level of context without sacrificing the safety required.

Digital Transformation

The adoption of digital technology by a company to improve business processes, value for customers, and innovation.

Digital Value Stream

A value stream is anything that delivers a product or a service. A digital value stream is one that delivers a digital product or service.

Dimensionality Reduction

Techniques for reducing the number of input variables in training data.

Distributed Debugging

The process of identifying the root cause of an unexpected behaviour in a distributed system.

Distributed Logging

Combines logs from many programs. It includes log aggregation.

Distributed Traces

Traces record the paths taken by requests (made by an application or end-user) as they propagate through multi-service architectures. It is a diagnostic technique that helps localize failures and performance issues within applications, especially those that may be distributed across multiple machines or processes. Tracing that spans multiple services and systems. Distributed tracing allows you to understand how requests flow through a complex system.

Distributed Version Control System (DVCS)

The software revisions are stored in a distributed revision control system (DRCS), also known as a distributed version control system (DVCS).

DMZ (De-Militarized Zone)

A DMZ in network security parlance is a network zone in between the public internet and internal protected resources. Any application, server, or service (including APIs) that need to be exposed externally are typically placed in a DMZ. It is not uncommon to have multiple DMZs in parallel.

Dojo

It is a hall or place for immersive learning, experiential learning, or meditation (Wikipedia).

Domain-Agnostic Solutions

Solutions focus is on solving problems across multiple domains. It operates more broadly and works across domains, monitoring, logging, cloud, infrastructure, and so on. These tools operate on vast amounts of IT data ingested from all domains/tools and they come up with models from this data to provide more accurate inferences and decisions.

Domain-Centric Solutions

Focused on solving a problem within a specific domain. Apply AIOps for a certain domain like network monitoring, log monitoring, application monitoring, or log collection.

DORA metrics

DevOps Research and Assessment (DORA) provides a standard set of DevOps metrics used for evaluating process performance and maturity. These metrics provide information about how quickly DevOps can respond to changes, the average time to deploy code, the frequency of iterations, and insight into failures.

Dynamic Analysis

Dynamic analysis is the testing of an application by executing data in real-time with the objective of detecting defects while it is in operation, rather than by repeatedly examining the code offline.

Dynamic Application Security Testing (DAST)

Dynamic application security testing (DAST) is a process of testing an application or software product in an operating state.

eBPF

Previously stood for Extended Berkeley Packet Filter (eBPF) which is a kernel technology that allows programs to run without requiring changes to the kernel source code or the addition of new modules. eBPF is useful in getting network sights and is immensely useful when instrumentation of tracing is not implemented across all components.

EggPlant

Automated function and regression testing of enterprise applications. Licensed by Test Plant.

Elasticity

In monitoring, elasticity is the ability of monitoring systems to scale up or down automatically with workload demand.

Elastic Infrastructure

Elasticity is a term typically used in cloud computing, to describe the ability of an IT infrastructure to quickly expand or cut back capacity and services without hindering or jeopardizing the infrastructure's stability, performance, security, governance, or compliance protocols.

ELK (Elasticsearch, Logstash, Kibana) Stack

An acronym used to describe a stack that comprises three popular projects: Elasticsearch, Logstash, and Kibana. All of the software components of ELK are free and open source.

Empiricism

It is the core philosophy that knowledge comes from experience and that decisions should be based on observed reality, facts, and data, rather than on upfront assumptions or rigid plans.

eNPS

Employee Net Promoter Score (eNPS) is a way for organizations to measure employee loyalty. The Net Promoter Score, originally a customer service tool, was later used internally on employees instead of customers.

Entity Under Test (EUT)

This is a class of terms that refers to the names of types of entities that are being tested. These terms are often abbreviated to the form xUT where "x" represents a type of entity under test.

Ephemeral Elastic Infrastructure

The concept of infrastructure being transitory, existing only briefly as needed to serve the needs of a DevOps process that needs infrastructure while it is executing.

Epics

Large bodies of work that represent significant features, initiatives, or objectives. They are broken down into smaller, manageable tasks called user stories.

Erickson (Stages of Psychosocial Development)

Erik Erikson (1950, 1963) proposed a psychoanalytic theory of psychosocial development comprising eight stages from infancy to adulthood. During each stage, the person experiences a psychosocial crisis which could have a positive or negative outcome for personality development.

Error Budget

The error budget provides a clear, objective metric that determines how unreliable a service is allowed to be within a specific time period.

Error Budget Policies

An error budget policy enumerates the activity a team takes when they've exhausted their error budget for a particular service in a particular time period.

Error Tracking

Tools to easily discover and show the errors that the application may be generating, along with the associated data.

Escalation Graph

A temporal graph, which consists of nodes and relations. The nodes of the escalation graph represent IT components at specific times.

Events

These are specific sequences of occurrences that take place within a system being monitored.

External Automation

Scripts and automation outside of a service that is intended to reduce toil.

Extended Detection and Response (XDR)

A unified security platform integrating multiple detection sources.

Extract, Transform, and Load (ETL)

The process of extracting data from source systems, transforming it into a usable format, and loading it into a target database or data warehouse.

Fail Early

A DevOps tenet referring to the preference to find critical problems as early as possible in a development and delivery pipeline.

Failure Mode

It refers to the different ways in which a system or component can fail or malfunction.

Fail Often

A DevOps tenet which emphasizes a preference to find critical problems as fast as possible and therefore frequently.

Failure Rate

Fail verdicts per unit of time.

False Negative

A test incorrectly reports a verdict of "fail" when the EUT actually passed the purpose of the test.

False Positive

A test incorrectly reports a verdict of "pass" when the EUT actually failed the purpose of the test.

Feature

In Agile methodologies, the features represent a chunk of functionality that delivers considerable business value, and that fulfills a stakeholder need. Features are a collection of user stories.

Feature Toggle

The practice of using software switches to hide or activate features. This enables continuous integration and testing a feature with selected stakeholders.

Federated Identity

A central identity used for access to a wide range of applications, systems, and services, but with a particular skew toward web-based applications. Also, often referenced as Identityas- a-Service (IDaaS). Any identity that can be reused across multiple sites, particularly via SAML or OAuth authentication mechanisms.

Filtering Metrics

These are quantifiable data points around the application and/or infrastructure. Example - spike in CPU usage, and so on.

Filtering Traces

It limits the events collected in a trace. If a filter is not set, all events of the selected event classes are returned in the trace output. It can optimize traces for lower cost in distributed cloud-native applications.

Fire Drills

A planned failure testing process focused on the operation of live services including service failure testing as well as communication, documentation, and other human factor testing.

Flow Engineering

It is defined as a structured approach that centers on a set of visual mapping exercises to analyze and improve the flow of value to customers. The Flow Engineering methodology uses Five Key Maps to achieve clarity, alignment, and fast flow.

FlipKart

A major Indian e-commerce company offering online retail and digital services.

Flow of Value

A form of map that shows the end-to-end value stream. This view is usually not available within the enterprise.

Flow of Work

It is the smooth, continuous movement of tasks, information, and deliverables across the software development and operations lifecycle. It encompasses the efficient progression of work from ideation to deployment and beyond, ensuring minimal delays, bottlenecks, or handoffs that could disrupt delivery.

Flow Metrics

A measure the flow of business value through all the activities involved in producing business value through a software value stream.

Flow Time

Flow time describes the full interval between the beginning and end of a particular process.

Framework

The backbone for plugging in tools. Launches automated tasks, collects results from automated tasks.

Freedom and Responsibility

A core cultural value that with the freedom of self-management (such as afforded by DevOps) comes the responsibility to be diligent, to follow the advice process, and to take ownership of both successes and failures.

Frequency

How often an application is released.

Functional Testing

Tests to determine if the functional operation of the service is as expected.

Future State Map

A form of value stream map that helps you develop and communicate what the target end state should look like and how to tackle the necessary changes.

Fuzzing

Fuzzing or fuzz testing is an automated software testing practice that inputs invalid, unexpected, or random data into applications.

Gated Commits

Define and obtain consensus for the criterion of changes promoted between all CD pipeline stages such as Dev to CI stage / CI to packaging/ delivery stage / Delivery to Deployment/ Production stage.

General Artificial Intelligence

It is also known as “strong AI”. This type of artificial intelligence that can solve unplanned, complex problems.

Generative (DevOps) Culture

In a generative organization, alignment takes place through identification with the mission. The individual ”buys into” what he or she is supposed to do and its effect on the outcome. Generative organizations tend to be proactive in getting the information to the right people by any means necessary (Westrum).

Generative AI

Algorithms (such as ChatGPT) that can be used to create new content, including audio, code, images, text, simulations, and videos.

Generativity

A cultural view wherein long-term outcomes are of primary focus, which in turn drives investments and cooperation that enable an organization to achieve those outcomes.

GIGO (Garbage In, Garbage Out)

Concept common to computer science and mathematics: the quality of output is determined by the quality of the input.

GitHub

A cloud platform for version control and collaborative software development using Git.

Glass-Box

Same as Clear-Box Testing and White-Box Testing.

Goal-seeking tests

The purpose of the test is to determine an EUT's performance boundaries, using incrementally stresses until the EUT reaches peak performance. For example, determininge the maximum throughput that can be handled without errors.

Golden Circle

A model by Simon Sinek that emphasizes an understanding of the business' "why" before focusing on the "what" and "how".

Golden Image

A template for a virtual machine (VM), virtual desktop, server, or hard disk drive. (TechTarget)

Goleman's Six Styles of Leadership

Daniel Goleman (2002) created the Six Leadership Styles and found, in his research, that leaders used one of these styles at any one time.

Google's Golden Signals

Four key metrics for monitoring services: latency, traffic, errors, and saturation.

Governance, Risk Management and Compliance (GRC)

A team or software platform intended for concentrating governance, compliance, and risk management data, including policies, compliance requirements, vulnerability data, and sometimes asset inventory, business continuity plans, and so on. In essence, a specialized document and data repository for security governance. Or a team of people who specialize in IT/security governance, risk management, and compliance activities. Most often non-technical business analyst resources.

Grafana

An open-source visualization and analytics platform for monitoring metrics, logs, and traces.

Gray-Box

Test cases use a limited knowledge of the internal design structure of the EUT.

gRPC

In tracing, gRPC is a high-performance remote procedure call framework whose calls can be instrumented for distributed tracing.

GUI testing

The purpose of the test is to determine if the graphical user interface operates as expected.

Guilds

A "community of interest" group that welcomes anyone and usually cuts across an entire organization. Similar to a Community of Practice.

Hadoop

An open-source framework for distributed storage and processing of large datasets.

Hand Offs

The procedure for transferring the responsibility of a particular task from one individual or team to another.

Hardening

Securing a server or infrastructure environment by removing or disabling unnecessary software, updating to known good versions of the operating system, restricting network-level access to only that which is needed, configuring logging in order to capture alerts, configuring appropriate access management, and installing appropriate security tools.

Helm Chart Registry

Helm charts are what describe related Kubernetes resources. Artifactory and Codefresh support a registry for maintaining master records of Helm Charts.

Heritage Reliability Engineer (HRE)

Applying the principles and practices of SRE to legacy applications and environments.

High-Trust Culture

Organizations with a high-trust culture encourage good information flow, crossfunctional collaboration, shared responsibilities, learning from failures and new ideas.

Hive

A data warehouse tool on Hadoop that provides SQL-like querying for big data.

Horizontal Scaling

Computing resources are scaled wider to increase the volume of processing. For example, adding more computers and running more tasks in parallel.

Hype cycle

A Gartner model showing the maturity, adoption, and expectations of emerging technologies.

Hyperscale

Hyperscale design allows architecture to scale as needed to meet increased demand by provisioning additional resources on demand to existing systems.

Hypothesis-Backlog

A collection of requirements expressed as experiments.

Hypothesis-Driven Development (HDD)

A prototype methodology that allows product designers to develop, test, and rebuild a product until it's acceptable to the users.

Idempotent

CM tools (for example, Puppet, Chef, Ansible, and Salt) claim that they are 'idempotent' by allowing the desired state of a server to be defined as code or declarations and automate steps necessary to consistently achieve the defined state time-after-time.

Identity

The unique name of a person, device, or the combination of both that is recognized by a digital system. Also referred to as an "account" or "user."

Identity and Access Management (IAM)

Policies, procedures, and tools for ensuring the right people have the right access to technology resources.

Identity as a Service (IDaaS)

Identity and access management services that are offered through the cloud or on a subscription basis.

Image-based test selection method

Build images are pre-assigned test cases. Tests cases are selected for a build by matching the image changes resulting from a build.

Immersion

A learning approach that guides teams with coaching and practice to help them learn to work in a new way.

Immutable

An immutable object is an object whose state cannot be modified after it is created. The antonym is a mutable object, which can be modified after it is created.

Immutable Infrastructures

Instead of instantiating an instance (server, container, and so on), with error-prone, timeconsuming patches and upgrades (that is mutations), replace it with another instance to introduce changes or ensure proper behaviour.

Impact-Driven Development (IDD)

A software development methodology that takes small steps towards achieving both impact and vision.

Impediment

Anything that prevents a team member from performing work as efficiently as possible.

Implementation Under Test

The EUT is a software implementation. For example, the embedded program is being tested.

Improvement Kata

A structured way to create a culture of continuous learning and improvement. (In Japanese business, Kata is the idea of doing things the "correct" way. An organization's culture can be characterized as its Kata through its consistent role modelling, teaching and coaching).

Incentive model

A system designed to motivate people to complete tasks toward achieving objectives. The system may employ either positive or negative consequences for motivation.

Incident

Any unplanned interruption to an IT service or reduction in the quality of an IT service. Includes events that disrupt or could disrupt the service. (ITIL definition)

Incident Management

A process that restores normal service operation as quickly as possible to minimize business impact and ensure that agreed levels of service quality are maintained. (ITIL definition) Involves capturing the who, what, when of service incidents and the onward use of this data in ensuring service level objectives are being met.

Incident Response

An organized approach to addressing and managing the aftermath of a security breach or attack (also known as an incident). The goal is to handle the situation in a way that limits damage and reduces recovery time and costs.

Increment

Potentially shippable completed work that is the outcome of a Sprint.

Incremental Rollout

Deploying many small, gradual changes to a service instead of a few large changes. Users are incrementally moved across to the new version of the service until eventually all users are moved across. Sometimes referred to by colored environments For example, blue/green deployment.

Inference

AIOps algorithms use the identified patterns, relationships, and historical data to make predictions and decisions. By applying machine learning models, AIOps can predict future outcomes or behaviours based on historical patterns and current data inputs.

InfluxDB

An open-source time-series database, optimized for metrics and events.

Infrastructure

All of the hardware, software, networks, facilities, and so on, required to develop, test, deliver, monitor and control or support IT services. The term IT infrastructure includes all of the information technology but not the associated people, processes, and documentation. (ITIL definition)

Infrastructure as Code (IaC)

The practice of using code (scripts) to configure and manage infrastructure.

Infrastructure Test

The purpose of the test is to verify the framework for EUT operating. For example, verifying specific operating system utilities function as expected in the target environment.

Infrastructure-as-a-Service (IaaS)

On-demand access to a shared pool of configurable computing resources.

Initiative

Initiatives, as used in the context of Agile, are collections of epics that drive toward a common goal.

Insights Driven

An insight-driven organization embeds analysis, data, and reasoning into the decision-making process, every day.

Instrumentation

Code that emits telemetry. The process of adding code to a system to collect data and provide observability. Instrumentation can include things like adding logging statements or implementing tracing.

Instrumenting with Agents

Any services used for monitoring. Agents sit outside the code, monitoring at the server or operating system level. It is “more complete” in the sense that it can work for pieces of the system whose code you don’t control, but less in-depth than in-app instrumentation with libraries.

Instrumenting with Libraries

Refers to the process of adding code to an application that allows it to emit information about its behaviour and performance. This information can include logs, metrics, and traces which can be used to monitor, diagnose, and resolve issues in real-time.

Integrated Development Environment (IDE)

An integrated development environment (IDE) is a software suite that consolidates the basic tools developers need to write and test software. Typically, an IDE contains a code editor, a compiler or interpreter, and a debugger that the developer accesses through a single graphical user interface (GUI). An IDE may be a standalone application, or it may be included as part of one or more existing and compatible applications. (TechTarget)

Integrated Development Environment (IDE) ‘lint’ checks

Linting is the process of running a program that will analyse code for potential errors (for example, formatting discrepancies, nonadherence to coding standards and conventions, logical errors).

Integration Test

It evaluates how different components or systems work together, ensuring that data flows correctly between modules, APIs, databases, and external services. These tests help confirm that combined parts function as expected in real-world scenarios.

Internet of Things

A network of physical devices that connect to the internet and potentially to each other through web-based wireless services.

Internal Automation

Scripts and automation delivered as part of the service that is intended to reduce toil.

Inverse of Conway’s Law

Organizing team structure to match the desired architecture of a system.

INVEST

A mnemonic was created by Bill Wake as a reminder of the characteristics of a quality user story.

ISO 31000

A family of standards that provide principles and generic guidelines on risk management.

Issue Management

A process for capturing, tracking, and resolving bugs and issues throughout the software development lifecycle.

IT Operations Analytics (ITOA)

It is an approach or method to retrieve, analyse, and report data for IT operations that leverages big analytics.

IT Service Management (ITSM)

Adopting a process approach towards management, focusing on customer needs and IT services for customers rather than IT systems, and stressing continual improvement. (Wikipedia)

iTest

Tool licensed by Spirent Communications for creating automated test cases.

ITIL (IT Infrastructure Library)

Provides a best practices framework that organizations can adapt to deliver and maintain IT services to provide optimal value for all stakeholders, including the customer.

IT Operations Management (ITOM)

The processes and tools for managing IT infrastructure and services.

Jaeger
An open-source distributed tracing system for monitoring microservices.

JavaScript Object Notation (JSON)
A lightweight format for data interchange.

Jenkins
Jenkins is a freeware tool. It is the most popular master automation framework tool, especially for continuous integration task automation. Jenkins task automation centres around timed processes. Many test tools and other tools offer plugins to simplify integration with Jenkins.

Kafka in tracing
Apache Kafka message streams instrumented to propagate and record distributed trace context.

Kaizen
The practice of continuous improvement.

Kanban
Method of work that pulls the flow of work through a process at a manageable pace.

Kanban Board
Tool that helps teams organize, visualize and manage work.

Karpman Drama Triangle
The drama triangle is a social model of human interaction. The triangle maps a type of destructive interaction that can occur between people in conflict.

Key Metrics
Something that is measured and reported upon to help manage a process, IT service or activity.

Key Performance Indicator (KPI)
Key performance indicators are the critical indicators of progress toward an intended result, providing a focus for improvement, and on what matters most.

Keywords-Based
Test cases are created using pre-defined names that reference programs useful for testing.

Knowledge Management
A process that ensures the right information is delivered to the right place or person at the right time to enable an informed decision.

Known Error
Problem with a documented root cause and a workaround. (ITIL definition)

Kolb’s Learning Styles
David Kolb published his learning styles model in 1984; his experiential learning theory works on two levels: a four-stage cycle of learning and four separate learning styles.

Kotter’s Dual Operating System
John Kotter describes the need for a dual operating system that combines the entrepreneurial capability of a network with the organizational efficiency of traditional hierarchy

Kubernetes
Kubernetes is an open-source container orchestration system for automating application deployment, scaling, and management. It was originally designed by Google and is now maintained by the Cloud Native Computing Foundation.

Kubernetes data plane
The components (pods, containers, services) that actually run workloads in Kubernetes clusters.

Kubler-Ross Change Curve
Describes and predicts the stages of personal and organizational reaction to major changes.

Lab-as-a-Service (LaaS)
Category of cloud computing services that provides a laboratory allowing customers to test applications without the complexity of building and maintaining the lab infrastructure.

Lagging Indicator
A lagging indicator is a statistical indicator that changes after conditions have already changed.

Laloux (Culture Models)
Frederic Laloux created a model for understanding organizational culture.

Large Language Model

A large language model is a language model consisting of a neural network with many parameters, trained on large quantities of unlabelled text using self-supervised learning or semi-supervised learning.

Latency

Latency is the delay incurred in communicating a message, the time a message spends 'on the wire' between the initial request being received, for example, by a server, and the response being received, for example, by a client.

Laws of Systems Thinking

In his book, 'The Fifth Discipline', Peter Senge outlines eleven laws that will help the understanding of business systems and to identify behaviours for addressing complex business problems.

Lead Time

Lead time measures the total time elapsed from the creation of work items to their completion. This is often confused with Cycle time which measures the time it takes for a team to complete work items once they begin actively working on them.

Leading Indicator

Leading indicators are sometimes described as inputs. They define what actions are necessary to achieve your goals with measurable outcomes.

Lean

Production philosophy that focuses on reducing waste and improving the flow of processes to improve overall customer value.

Lean (adjective)

Spare, economical, lacking richness or abundance.

Lean Canvas

Lean Canvas is a 1-page business plan template.

Lean Enterprise

An organization that strategically applies the key ideas behind lean production across the enterprise.

Lean IT

Applying the key ideas behind lean production to the development and management of IT products and services.

Lean Manufacturing

Lean production philosophy derived mostly from the Toyota Production System.

Lean Product Development

Lean Product Development, or LPD, utilizes Lean principles to meet the challenges of Product Development.

Lean Startup

A system for developing a business or product in the most efficient way possible to reduce the risk of failure.

Lean Thinking

It aims to maximize customer value while minimizing waste by optimizing resources and processes. Waste refers to any activity that does not add value to the final product or service.

Learning Organization

An organization that fosters a culture of continuous learning and improvement, recognizing that innovation and long-term success depend on the ability to acquire, apply, and adapt knowledge. This commitment to learning is embedded into daily practices, ensuring that individuals and teams develop new skills, avoid cultural debt, and enhance organizational resilience.

License Scanning

Tools, such as Blackduck and Synopsis, that check that licenses of your dependencies are compatible with your application, and approve or blacklist them.

Litmus Chaos

A Kubernetes-native Chaos Engineering framework for resilience testing.

Little's Law

A theorem by John Little that states that the long-term average number L of customers in a stationary system is equal to the long-term average effective arrival rate λ multiplied by the average time W that a customer spends in the system.

Load Balancer

A system that distributes network or application traffic across multiple servers.

LoadRunner

A tool used to test applications, measuring system behaviour, and performance under load. Licensed by HP.

Local Development Scanning

Security scanning here is limited to the tools used by developers to scan code when creating or modifying container images. This practice should be a "must" to avoid future security issues in the Continuous Integration process. This is the more proactive moment to apply a security scanning.

Logs

Serialized report of details such as test activities and EUT console logs. A record of events that have occurred within a system. Logs can be used to troubleshoot issues and provide insight into how a system is behaving.

Log Analysis

Tools that allow searches, count, and sometimes correlate data from logs.

Log Management

The collective processes and policies used to administer and facilitate the generation, transmission, analysis, storage, archiving, and ultimate disposal of the large volumes of log data created within an information system.

Logging

The capture, aggregation, and storage of all logs associated with system performance including, but not limited to, process calls, events, user data, responses, error, and status codes. Logstash and Nagios are popular examples.

Logic Bomb (Slag Code)

A string of malicious code used to cause harm to a system when the programmed conditions are met.

Longevity Test

The purpose of the test is to determine if a complete system performs as expected over an extended period of time.

Machine Learning

Data analysis that uses algorithms to recognize patterns from large amounts of data to learn from that data.

Machine Learning Models (ML Models)

It is a program that can find patterns or make decisions from a previously unseen dataset.

Maintainability

It is how quickly and easily can something be fixed.

Malware

A program designed to gain access to computer systems, normally for the benefit of some third party, without the user's permission.

Many-factor Authentication

The practice of using at least 2 factors for authentication. The two factors can be of the same class.

Mean-Time Between Deploys

Used to measure deployment frequency.

Mean-Time Between Failures (MTBF)

The average time that a CI or IT service can perform its agreed function without interruption. Often used to measure reliability. Measured from when the CI or service starts working, until the time it fails (uptime). (ITIL definition)

Mean-Time to Detect Defects (MTTD)

Average time required to detect a failed component or device.

Mean-Time to Discovery

How long a vulnerability or software bug/defect exists before it's identified.

Mean-Time to Patch

How long it takes to apply patches to environments once a vulnerability has been identified.

Mean-Time to Repair/Recover (MTTR)

Average time required to repair/recover a failed component or device. MTTR does not include the time required to recover or restore service.

Mean-Time to Restore Service (MTRS)

Used to measure time from when the CI or IT service fails until it is fully restored and delivering its normal functionality (downtime). Often used to measure maintainability. (ITIL definition)

MELT

An acronym that stands for Metrics, Events, Logs, and Traces. These are the essential data types for Observability. When we instrument everything and use this acronym to form a fundamental, working knowledge of connections (the relationships and dependencies within our system) as well as its detailed performance and health, exemplifies Observability.

Mental Models

A mental model is an explanation of someone's thought process about how something works in the real world.

Merge

The action of integrating software changes together into a software version management system.

Metric

Something that is measured and reported upon to help manage a process, IT service, or activity. Quantitative measurements that provide insight into how a system is behaving. Metrics can include things like CPU usage, memory usage, network traffic, and error rates.

Metrics

This is a class of terms relevant to measurements used to monitor the health of a product or infrastructure.

Microprocess

A distinct activity that can be defined, designed, implemented, and managed independently and is generally associated with a primary service management practice. A microprocess may be integrated with other service management practices.

Microprocess Architecture

A collection of integrated microprocesses that collectively perform all of the activities necessary for an end-to-end service management practice to be successful.

Microservices

A software architecture that is composed of smaller modules that interact through APIs and can be updated without affecting the entire system.

Mindset

A person's usual attitude or mental state is their mindset.

Minimum Viable Process

The least amount needed in order for this process or microprocess to meet its Definition of Done.

Minimum Viable Product

Most minimal version of a product that can be released and still provide enough value that people are willing to use it.

MLOps

MLOps or ML Ops is a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently. The word is a compound of "machine learning" and the continuous development practice of DevOps in the software field.

Mock Object

Mock is a method/object that simulates the behavior of a real method/object in controlled ways. Mock objects are used in unit testing. Often a method under a test calls other external services or methods within it. These are called dependencies.

Model

Representation of a system, process, IT service, CI, and so on, that is used to help understand or predict future behaviour. In the context of processes, models represent pre-defined steps for handling specific types of transactions.

Model-Based

Test cases are automatically derived from a model of the entity under test. Example tool: Tricentis.

Monitoring

The use of a hardware or software component to monitor the system resources and performance of a computer service.

Monitoring Tools

Tools that allow IT organizations to identify specific issues of specific releases and to understand the impact on end-users.

Monolithic

A software system is called "monolithic" if it has a monolithic architecture, in which functionally distinguishable aspects (for example data input and output, data processing, error handling, and the user interface) are all interwoven, rather than containing architecturally separate components.

Multi-factor Authentication

The practice of using 2 or more factors for authentication. Often used synonymously with 2-factor Authentication.

Multi-cloud

Multi-cloud DevOps solutions provide ondemand multi-tenant access to development and test environments.

Nagios

An open-source IT infrastructure monitoring and alerting tool.

Narrow Artificial Intelligence (AI)

It is also known as "weak AI." This is a taskspecific type of artificial intelligence in which a learning algorithm is designed to perform a single task. It has very limited parameters.

Natural Language Processing (NLP)

AI techniques for enabling computers to understand human language.

Network Observability

The ability to understand, analyse, and troubleshoot network behaviour using Telemetry.

Network Reliability Engineer (NRE)

Someone who applies a reliability engineering approach to measure and automate the reliability of networks.

Network Virtual Appliances

Software-based appliances that deliver networking functions traditionally handled by hardware.

Neural networks

A computational model inspired by the human brain, consisting of interconnected nodes (neurons) that process data in layers to recognize patterns and make predictions.

Neuroplasticity

Describes the ability of the brain to form and reorganize synaptic connections, especially in response to learning or experience or following injury.

Neuroscience

The study of the brain and nervous system.

Non-functional requirements

Requirements that specify criteria that can be used to judge the operation of a system, rather than specific behaviours or functions (for example, availability, reliability, maintainability, supportability); qualities of a system.

Non-functional tests

Defined as a type of service testing intending to check non-functional aspects such as performance, usability, and reliability of a software service.

Object Under Test (OUT)

The EUT is a software object or class of objects.

Observability

Observability is focused on externalizing as much data as you can about the whole service allowing us to infer what the current state of that service from its external behaviour.

Observability Backend

The data access layer to include storage and UI (user interface). There are three main components that the backend is responsible for: to include data storage, query service and visualization.

Observability Center of Excellence

The highest level of observability patterns within an organization with a unit that focuses on best practices and standards. Support, education, and training is also provided to individuals and teams within the organization.

Observability-Driven Design (ODD)

Enables Observability of a system throughout the entire development cycle; it's about encouraging developers to wrap their code with breadcrumbs that can be traced and increase logging and events at the coding level. It's an approach to shift left observability to the earliest stage of the software development life cycle.

Observability Enablement

An introductory level of observability patterns within an organization controlled by experienced DevOps Engineers or SREs. Usually, this enablement process collects and documents best practices along with tool management guidance.

Observability lake

A centralized repository storing metrics, logs, and traces for large-scale Observability analysis.

Observability Maturity Model

A model that defines four distinct levels in the evolution of observability.

Objectives and Key Results (OKRs)

Objectives and key results is a goal-setting framework used by individuals, teams, and organizations to define measurable goals and track their outcomes.

On-call

Being on-call means someone being available during a set period of time, and being ready to respond to production incidents during that time with appropriate urgency.

OODA loop (Observe, Orient, Decide, Act)

A continuous decision-making cycle that emphasizes rapidly observing a situation, orienting to context, deciding on a course of action, and acting; then repeating the loop to adapt to change.

OpenCensus standard

An open-source Observability framework for metrics and distributed tracing (now merged into OpenTelemetry).

Open Source

Software that is distributed with its source code so that end-user organizations and vendors can modify it for their own purposes.

OpenTelemetry (OTel)

The de facto open standard for instrumentation. It is the project that provides a set of APIs, SDKs, libraries, and agents to enable developers to instrument their application once and send correlated traces and metrics to a destination of their choice. It supports 11 languages (so far) and automatic instrumentation agents, including Java, Javascript, Python, .NET, and more.

Open Telemetry Protocol (OTLP)

A general-purpose telemetry data delivery protocol designed in the scope of OpenTelemetry project. The specification describes the encoding, transport, and delivery mechanism of telemetry data between telemetry sources, intermediate nodes such as collectors and telemetry backends.

OpenTracing standard

A vendor-neutral API standard for distributed tracing.

OpenTSDB

An open-source time-series database built on HBase for scalable metric storage.

Operations (Ops)

Individuals involved in the daily operational activities needed to deploy and manage systems and services such as quality assurance analysts, release managers, system and network administrators, information security officers, IT operations specialists, and service desk analysts.

Operations Management

The function that performs the daily activities needed to deliver and support IT services and the supporting IT infrastructure at the agreed levels. (ITIL)

Orchestration

An approach to building automation that interfaces or "orchestrates" multiple tools together to form a toolchain.

Organization Culture

A system of shared values, assumptions, beliefs, and norms that unite the members of an organization.

Organization Model
For DevOps, an approach that is not a dominator hierarchy but instead a Distributed Autonomous Organization (DAO).

Organizational Change
Efforts to adapt the behaviour of humans within an organization to meet new structures, processes, or requirements.

OS Virtualization
A method for splitting a server into multiple partitions called "containers" or "virtual environments" in order to prevent applications from interfering with each other.

OTel specification
OpenTelemetry specification defining standards for collecting and exporting Telemetry data.

Outcome
Intended or actual results.

Outcome Mapping
A methodology for planning, monitoring, and evaluating development initiatives in order to bring about sustainable change.

Package Registry
A repository for software packages, artifacts, and their corresponding metadata. Can store files produced by an organization itself or for third-party binaries. Artifactory and Nexus are amongst the most popular.

Pages
Something for creating supporting web pages automatically as part of a CI/CD pipeline.

Patch
A software update designed to address (mitigate/remediate) a bug or weakness.

Patch management
The process of identifying and implementing patches.

Pathological Culture
Pathological cultures tend to view information as a personal resource, to be used in political power struggles. (Westrum)

Penetration Testing
An authorized simulated attack on a computer system that looks for security weaknesses, potentially gaining access to the system's features and data.

People Changes
Focuses on changing attitudes, behaviours, skills, or performance of employees.

Performance Test
The purpose of the test is to determine an EUT meets its system performance criterion or to determine what a system's performance capabilities are.

Plan-Do-Check-Act
A four-stage cycle for process management and improvement attributed to W. Edwards Deming. Sometimes called the Deming Cycle or PDCA.

Platform-as-a-Service (PaaS)
Category of cloud computing services that provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure.

Platform Engineering
Platform engineering is a software engineering discipline that focuses on building toolchains and self-service workflows for the use of developers. Platform engineering is about creating a shared platform for software engineers using computer code.

Plugin
A pre-programmed integration between an orchestration tool and other tools. For example, many tools offer plugins to integrate with Jenkins.

Policies
Formal documents that define boundaries in terms of what the organization may or may not do as part of its operations.

Policy as Code
The notion that security principles and concepts can be articulated in code (for example, software, configuration management, automation) to a sufficient degree that the need for an extensive traditional policy

framework is greatly reduced. Standards and guidelines should be implemented in code and configuration, automatically enforced, and automatically reported in terms of compliance, variance, or suspected violations.

Practice

A complete end-to-end capability for managing a specific aspect of service delivery (for example, changes, incidents, service levels).

Practice Backlog

A prioritized list of everything that needs to be designed or improved for a practice including current and future requirements.

Practice/Microprocess Planning

A high-level event to define the goals, objectives, inputs, outcomes, activities, stakeholders, tools, and other aspects of a practice or microprocess. This meeting is not timeboxed.

Predictive Analytics

A branch of advanced analytics that makes predictions about future outcomes. It can be used to mine performance data to identify potential problems. This insight can provide guidance to SRE.

Pre-Flight

This is a class of terms that refers to names of activities and processes that are conducted on an EUT prior to integration into the trunk branch.

Priority

The relative importance of an incident, problem, or change; based on impact and urgency. (ITIL definition)

Privileged Access Management (PAM)

Technologies that help organizations provide secured privileged access to critical assets and meet compliance requirements by securing, managing, and monitoring privileged accounts and access. (Gartner)

Proactive Approach

It is an approach that attempts to understand a system even before it fails (unacceptable quality) in an attempt to identify how it could fail in the future.

Problem

The underlying cause of one or more incidents. (ITIL definition)

Probabilistic Systems

Systems in which the outcome or behaviour is influenced by randomness or uncertainty.

Process

A structured set of activities designed to accomplish a specific objective. A process takes inputs and turns them into defined outputs. Related work activities that take specific inputs and produce specific outputs that are of value to a customer.

Process Changes

Focuses on changes to standard IT processes, such as software development practices, ITIL processes, change management, approvals, and so on.

Process Owner

A role accountable for the overall quality of a process. It may be assigned to the same person who carries out the Process Manager role, but the two roles may be separate in larger organizations. (ITIL definition)

Process Standup

A time-boxed event of 15 minutes to inspect progress towards the Sprint Goal and identify impediments as quickly as possible.

Processing Time

The period during which one or more inputs are transformed into a finished product by a manufacturing or development procedure. (Business Dictionary)

Product Backlog

Prioritized list of functional and non-functional requirements for a system usually expressed as user stories.

Product Owner

An individual responsible for maximizing the value of a product and for managing the product backlog. Prioritizes, grooms, and owns the backlog. Gives the squad purpose.

Programming-Based
Test cases are created by writing code in a programming language. For example, JavaScript, Python, TCL, Ruby.

Project to Product
Changing ways of working from a large batch, waterfall project led approach, to a small batch, agile product (or value stream) approach.

Prometheus
An open-source monitoring and alerting system focused on time-series metrics.

Provision Platforms
Tools that provide platforms for provisioning infrastructure (for example, Puppet, Chef, Salt).

Psychological Safety
Psychological safety is a shared belief that the team is safe for interpersonal risk-taking.

QTP
Quick Test Professional is a functional and regression test automation tool for software applications. Licensed by HP.

Quality Management
Tools that handle test case planning, test execution, defect tracking (often into backlogs), severity, and priority analysis. CA's Agile Central.

RAMS
It stands for Reliability, Availability, Maintainability, and Safety, which are key considerations in the design and operation of complex systems.

Ranorex
GUI test automation framework for testing of desktop, web-based and mobile applications. Licensed by Ranorex.

Ransomware
Encrypts the files on a user's device or a network's storage devices. To restore access to the encrypted files, the user must pay a "ransom" to the cybercriminals, typically through a tough-to-trace electronic payment method such as Bitcoin.

RASP
Runtime Application Self-Protection.

Reactive Approach
It is characterized by addressing issues and incidents as they occur, without proactive measures or preventive actions.

RED (Rate, Errors, Duration) Metrics View
A monitoring acronym that stands for rate, errors and duration. It is a subset of the golden signals but provides basic steps for monitoring web applications.

Registry Scanning
It's required that scanning tools are connected to the container registry or provided by itself. It's an important step to observe security vulnerabilities that could be found in container Images and in other Cloud Native artifacts.

Regression model
In AI, regression model is a supervised learning approach that predicts a continuous numerical outcome based on relationships between input features and target values.

Regression testing
The purpose of the test is to determine if a new version of an EUT has broken some things that worked previously.

Regulatory compliance testing
The purpose of the test is to determine if an EUT conforms to specific regulatory requirements. For example, verifying that an EUT satisfies government regulations for consumer credit card processing.

Release
Software that is built, tested, and ready to be deployed into the production environment.

Release Acceptance Criteria
Measurable attributes for a release package that determine whether a release candidate is acceptable for deployment to customers.

Release Candidate
A release package that has been prepared for deployment, may or may not have passed the Release.

Release Governance

Release Governance is all about the controls and automation (security, compliance, or otherwise) that ensure your releases are managed in an auditable and trackable way, in order to meet the need of the business to understand what is changing

Release Management

The process that manages releases and underpins Continuous Delivery and the Deployment Pipeline.

Release Orchestration

Typically a deployment pipeline used to detect any changes that will lead to problems in production. Orchestrating other tools will identify performance, security, or usability issues. Tools like Jenkins and Gitlab CI can "orchestrate" releases.

Relevance

A Continuous Testing tenet which emphasizes a preference to focus on the most important tests and test results.

Reliability

A measure of how long a service, component, or CI can perform its agreed function without interruption. Usually measured as MTBF or MTBSI. (ITIL definition)

Reliability Test

The purpose of the test is to determine if a complete system performs as expected under stressful and loaded conditions over an extended period of time.

Remediation

Action of addressing and resolving a problem, IT issues, incidents, or anomalies found during DevOps processes or by an AIOps system. For example, roll-back changes for an EUT change that resulted in a CT test case fail verdict.

Remediation Plan

A plan that determines the actions to take after a failed change or release. (ITIL definition)

Request for Change (RFC)

Formal proposal to make a change. The term RFC is often misused to mean a change record, or the change itself. (ITIL definition)

Requirements Management

Tools that handle requirements definition, traceability, hierarchies & dependency. Often also handles code requirements and test cases for requirements.

Resilience

Building an environment or organization that is tolerant to change and incidents.

Resilience engineering

The study and practice of designing systems and organizations to adapt, recover, and thrive in the face of unexpected challenges, disruptions, or changes.

Response Time

Response time is the total time it takes from when a user makes a request until they receive a response.

REST

Representation State Transfer. The software architecture style of the worldwide web.

Restful API

Representational state transfer (REST) or RESTful services on a network, such as HTTP, provide scalable interoperability for requesting systems to quickly and reliably access and manipulate textual representations (XML, HTML, JSON) of resources using stateless operations (GET, POST, PUT, DELETE, and so on.).

RESTful interface testing

The purpose of the test is to determine if an API satisfies its design criterion and the expectations of the REST architecture.

Return on Investment (ROI)

The difference between the benefit achieved and the cost to achieve that benefit, expressed as a percentage.

Review Apps

Allow code to be committed and launched in real-time – environments are spun up to allow developers to review their application.

Rework

The time and effort required to correct defects (waste).

Risk
A possible event that could cause harm or loss or affect an organization's ability to achieve its objectives. The management of risk consists of three activities: identifying risks, analysing risks, and managing risks. The probable frequency and probable magnitude of future loss. Pertains to a possible event that could cause harm or loss or affect an organization's ability to execute or achieve its objectives.

Risk Event
A possible event that could cause harm or loss or affect an organization's ability to achieve its objectives. The management of risk consists of three activities: identifying risks, analysing risks, and managing risks.

Risk Management Process
The process by which "risk" is contextualized, assessed and treated. From ISO 31000: 1) Establish context, 2) Assess risk, 3) Treat risk (remediate, reduce or accept).

Robot Framework
TDD framework created and supported by Google.

Role
Set of responsibilities, activities, and authorities granted to a person or team. A role is defined by a process. One person or team may have multiple roles. A set of permissions assigned to a user or group of users to allow a user to perform actions within a system or application.

Role-based Access Control (RBAC)
An approach to restricting system access to authorized users.

Roll-back
Software changes which have been integrated are removed from the integration.

Root Cause Analysis (RCA)
Actions taken to identify the underlying cause of a problem or incident.

Root Span
Created to represent the whole activity. It is the only span without a parent and represents how a service request is started, and when the response is sent.

Rugged Development (DevOps)
Rugged Development (DevOps) is a method that includes security practices as early in the continuous delivery pipeline as possible to increase cybersecurity, speed, and quality of releases beyond what DevOps practices can yield alone.

Rugged DevOps
Rugged DevOps is a method that includes security practices as early in the continuous delivery pipeline as possible to increase cybersecurity, speed, and quality of releases beyond what DevOps practices can yield alone.

Runbooks
A collection of procedures necessary for the smooth operation of a service. Previously manual in nature they are now usually automated with tools like Ansible.

Runtime Application Self Protection (RASP)
Tools that actively monitor and block threats in the production environment before they can exploit vulnerabilities.

Sample Rate
The number of spans represented by this particular span, when some portion of spans are dropped to save storage. This makes spans and traces meaningfully countable without keeping each one.

Sampling
The concept of selecting a few elements from a large collection and learning about the entire collection by extrapolating from the selected set. It helps to reduce the volume of tracing data.

Sanity Test
A very basic set of tests that determine if a software is functional at all.

SANS
SysAdmin, Audit, Network, and Security Institute, a cybersecurity training and certification organization.

Saturation
The degree to which the resource has extra work which it can't service, often queued (backlog).

Scalability

Scalability is a characteristic of a service that describes its capability to cope and perform under an increased or expanding load.

Scaled Agile Framework (SAFe)

A proven, publicly available, framework for applying Lean-Agile principles and practices at an enterprise scale.

SCARF Model

A summary of important discoveries from neuroscience about the way people interact socially.

Scheduling

Scheduling: the process of planning to release changes into production.

Scrum

A simple framework for effective team collaboration on complex projects. Scrum provides a small set of rules that create "just enough" structure for teams to be able to focus their innovation on solving what might otherwise be an insurmountable challenge. (Scrum.org)

Scrum Pillars

Pillars that uphold the Scrum framework include Transparency, Inspection, and Adaption.

Scrum Team

A self-organizing, cross-functional team that uses the Scrum framework to deliver products iteratively and incrementally. The Scrum Team consists of a Product Owner, Developers, and a Scrum Master.

Scrum Values

A set of fundamental values and qualities underpinning the Scrum framework: commitment, focus, openness, respect and courage.

Scrum Master

An individual who provides process leadership for Scrum (that is, it ensures Scrum practices are understood and followed) and who supports the Scrum Team by removing impediments.

Secret Detection

Secret Detection aims to prevent that sensitive information, like passwords, authentication tokens, and private keys are unintentionally leaked as part of the repository content.

Secrets Management

Secrets management refers to the tools and methods for managing digital authentication credentials (secrets), including passwords, keys, APIs, and tokens for use in applications, services, privileged accounts, and other sensitive parts of the IT ecosystem.

Secure Automation

Secure automation removes the chance of human error (and wilful sabotage) by securing the tooling used across the delivery pipeline.

Security (Information Security)

Practices intended to protect the confidentiality, integrity, and availability of computer system data from those with malicious intentions.

Security as Code

Automating and building security into DevOps tools and practices, making it an essential part of toolchains and workflows.

Security Information and Event Management (SIEM)

Systems that analyse and correlate security data in real time.

Security Orchestration, Automation, and Response (SOAR)

Platforms that automate and coordinate security workflows.

Security tests

The purpose of the test is to determine if an EUT meets its security requirements. An example is a test that determines if an EUT processes login credentials properly.

Selenium

Popular open-source tool for software testing GUI and web applications.

Self-healing

Self-healing means the ability of services and underlying environments to detect and resolve problems automatically. It eliminates the need for manual human intervention.

Semi-Structured Data

This type of data has some elements of structure as it contains metadata and tags but does not conform to a specific data model. This data cannot be stored in rows and columns or databases.

Serverless

A code execution paradigm where no underlying infrastructure or dependencies are needed, moreover, a piece of code is executed by a service provider (typically cloud) who takes over the creation of the execution environment. Lambda functions in AWS and Azure Functions are examples.

Service

Enables the ability to do something when and how it is needed or desired. It enables its customers to achieve their objectives more efficiently and/or more effectively than they could without the service.

Service Architecture

Mechanisms to enable access to one or more capabilities, where the access is provided using a prescribed interface and is exercised. (Wikipedia)

Service Desk

Single point of contact between the service provider and the users. Tools like Service Now are used for managing the lifecycle of services as well as internal and external stakeholder engagement.

Service Level Agreement (SLA)

Written agreement between an IT service provider and its customer(s) that defines key service targets and responsibilities of both parties. An SLA may cover multiple services or customers. (ITIL definition)

Service Level Indicator (SLI)

SLI's are used to communicate quantitative data about services, typically to measure how the service is performing against an SLO.

Service Level Objective (SLO)

An SLO is a goal for how well a product or service should operate. SLO's are set based on what an organization is expecting from a service.

Service Map

Used to discover and track IT applications and components in an organization, outlining how each connects to business services.

Seven Pillars of DevOps

Seven distinct "pillars" provide a foundation for DevOps systems which include Collaborative Culture, Design for DevOps, Continuous Integration, Continuous Testing, Continuous Delivery and Deployment, Continuous Monitoring, and Elastic Infrastructure and Tools.

Shift Left

An approach that strives to build quality into the software development process by incorporating testing early and often. This notion extends to security architecture, hardening images, application security testing, and beyond.

Signals

Forms of communication that are emitted from traces, logs, events and metrics.

SilkTest

Automated function and regression testing of enterprise applications. Licensed by Borland.

Simian Army

The Simian Army is a suite of failure-inducing tools designed by Netflix. The most famous example is Chaos Monkey which randomly terminates services in production as part of a Chaos Engineering approach.

Simple Network Management Protocol (SNMP)

A standard protocol for managing and monitoring network devices.

Single Point of Failure (SPOF)

A single point of failure (SPOF) is a part of a system that, if it fails, will stop the entire system from working.

Site Reliability Engineering (SRE)

The discipline that incorporates aspects of software engineering and applies them to infrastructure and operations problems. The main goals are to create scalable and highly reliable software systems.

Smoke Test

A basic set of functional tests that are run immediately after a software component is built. Same as CI Regression Test.

Snapshot

Report of pass/fail results for a specific build.

Snippets

Stored and shared code snippets to allow collaboration around specific pieces of code. Also allows code snippets to be used in other code-bases. BitBucket and GitLab allow this.

SOAP

Simple Object Access Protocol (SOAP) is an XML-based messaging protocol for exchanging information among computers.

Software Composition Analysis

A tool that checks for libraries or functions in source code that have known vulnerabilities.

Software Defined Networking (SDN)

Software-Defined Networking (SDN) is a network architecture approach that enables the network to be intelligently and centrally controlled, or 'programmed,' using software applications.

Software Delivery Lifecycle (SDLC)

The process used to design, develop and test high quality software.

Software Version Management System

A repository tool which is used to manage software changes. Examples are: Azure DevOps, BitBucket, Git, GitHub, GitLab, VSTS.

Software-as-a-Service (SaaS)

Category of cloud computing services in which software is licensed on a subscription basis.

Source Code Tools

Repositories for controlling source code for key assets (application and infrastructure) as a single source of truth.

Span

A single operation with a name, timestamp, duration, tags, and other metadata. It represents a single unit of work within a trace.

Span events

In tracing, span events are detailed logs attached to spans that record granular execution events.

Span ID

A short universally unique identifier (UUID) to identify a span, usually 16 hex characters.

Spotify Squad Model

An organizational model that helps teams in large organizations behave like startups and be nimble.

Sprint

A period of 2-4 weeks during which an increment of product work is completed.

Sprint (Scrum)

A time-boxed iteration of work during which an increment of product functionality is implemented.

Sprint Backlog

Subset of the backlog that represents the work that must be completed to realize the Sprint Goal.

Sprint Goal

The purpose and objective of a Sprint, often expressed as a business problem that is going to be solved.

Sprint Planning

A 4 to 8-hour time-boxed event that defines the Sprint Goal, the increment of the Practice Backlog that will be completed during the Sprint, and how it will be completed.

Sprint Retrospective

A 1.5 to 3-hour time-boxed event during which the Team reviews the last Sprint and identifies and prioritizes improvements for the next Sprint.

Sprint Review

A time-boxed event of four hours or less where the Team and stakeholders inspect the work resulting from the Sprint and update the Practice Backlog.

Spyware
Software that is installed in a computer without the user's knowledge and transmits information about the user's computer activities over back to the threat agent.

Squads
A cross-functional, co-located, autonomous, self-directed team.

SSH
Secure Shell, a protocol for encrypted remote access and administration of systems.

Stakeholder
Person who has an interest in an organization, project or IT service. Stakeholders may include customers, users and suppliers

Stability
The sensitivity a service has to accept changes and the negative impact that may be caused by system changes. Services may have reliability, in that if functions over a long period of time, but may not be easy to change and so does not have stability.

Standard Change
Pre-approved, low risk change that follows a procedure or work instruction. (ITIL definition)

Static Application Security Testing (SAST)
A type of testing that checks source code for bugs and weaknesses.

Static Code Analysis
The purpose of the test is to detect source code logic errors and omissions such as memory leaks, unutilized variables, unutilized pointers.

StatsD
A network daemon for collecting, aggregating, and exporting application metrics.

Status Page
Service pages that easily communicate the status of services to customers and users.

Sticks
Negative incentives, for discouraging or punishing undesired behaviours.

Storage Security
A specialty area of security that is concerned with securing data storage systems and ecosystems and the data that resides on these systems.

Stormstack
A commercial orchestration tool based on event triggers instead of time-based.

StoStaKee
This stands for stop, start, and keep: this is an interactive time-boxed exercise focused on past events.

Strategic Sprint
A less than a week timeboxed Sprint during which strategic elements that were defined during Practice Planning are completed so that the Team can move on to designing the activities of the process.

Stream-Aligned Team
A team aligned to a single, valuable stream of work; this might be a single product or service, a single user story, or a single user persona.

Streaming Data
Streaming data is data that is continuously generated by different sources. Such data should be processed incrementally using stream processing techniques without having access to all of the data.

Structural Changes
Changes in the hierarchy of authority, goals, structural characteristics, administrative procedures, and management systems.

Structured Data
It is when data is in a standardized format with a well-defined structure and is easily accessed by humans and programs. This data type is generally stored in a database.

Structured Logs
Forms of logs that are computer-readable with many attributes (not just messages).

Super Artificial Intelligence

Super artificial intelligence (AI) is a form of AI that has the ability to surpass human abilities and performance by manifesting cognitive skills and developing thinking skills of its own.

Supervised learning

A type of Machine Learning where an algorithm learns from labelled training data to make predictions or classifications.

Supplier

External (third party) supplier, manufacturer, or vendor responsible for supplying goods or services that are required to deliver IT services.

Synthetic Monitoring

Synthetic monitoring (also known as active monitoring, or semantic monitoring) runs a subset of an application's automated tests against the system on a regular basis. The results are pushed into the monitoring service, which triggers alerts in case of failures.

Syslog

A standard protocol for logging system and network events.

System of Record

A system of record is the authoritative data source for a data element or data entity.

System Test

The purpose of the test is to determine if a complete system performs as expected in its intended configurations.

System Under Test (SUT)

The EUT is an entire system. For example, Bank teller machine is being tested.

Systems Thinking

A core concept in VSM, by emphasizing that efficiency can only be achieved by looking at the interconnected nature of the entire organization and process, rather than isolating individual departments or tasks.

Tag-Based Test Selection Method

Tests and Code modules are pre-assigned tags. Tests are selected for a build matching preassigned tags.

Takt Time

Takt time is the rate at which you need to complete a product in order to meet customer demand. It comes from the German word "Takt," meaning beat or pulse in music. Within manufacturing, takt is an important measure of output against demand.

Target Operating Model

A description of the desired state of the operating model of an organization.

Teal Organization

An emerging organizational paradigm that advocates a level of consciousness including all previous world views within the operations of an organization.

Team Dynamics

A measurement of how a team works together. Includes team culture, communication styles, decision-making ability, trust between members, and the willingness of the team to change.

Team Topologies

An approach to organizing business and technology teams for fast flow, providing a practical, step-by-step, adaptive model for organizational design and team interaction.

Techno-Economic Paradigm Shifts

Techno-economic paradigm shifts are at the core of the general, innovation-based theory of economic and societal development as conceived by Carlota Perez.

Telemetry

Telemetry is the collection of measurements or other data at remote or inaccessible points and their automatic transmission to receiving equipment for monitoring.

Temporal alerts

Alerts triggered based on conditions persisting over a specified period of time.

Temporal Relations

Describes timelines for each component. It always only connects components with their past self each time a significant event occurs (in terms of the problem).

Terraform

An infrastructure-as-code tool for provisioning and managing cloud resources declaratively.

Test Architect

Person who has responsibility for defining the overall end-to-end test strategy for an EUT.

Test Artifact Repository

Database of files used for testing.

Test Campaign

A test campaign may include one or more test sessions.

Test Case

Set of test steps together with data and configuration information. A test case has a specific purpose to test at least one attribute of the EUT.

Test Creation Methods

This is a class of test terms that refers to the methodology used to create test cases.

Test-Driven Development (TDD)

Test-driven development (TDD) is a software development process in which the developer writes a test before composing code. They then follow this process:

1. Write the test.
2. Run the test and any others that are relevant and see them fail.
3. Write the code.
4. Run test(s).
5. Refactor code if needed.
6. Repeat.

Unit level tests and/or application tests are created ahead of the code that is to be tested.

Test Duration

The time it takes to run a test. For example, number of hours per test.

Test Environment

The test environment refers to the operating system (for example, Linus, windows version, and so on), the configuration of software (for example, parameter options), dynamic conditions (for example, CPU and memory utilization), and physical environment (for example, power, cooling) in which the tests are performed.

Test Fast

A CT tenet referring to accelerated testing.

Test Framework

A set of processes, procedures, abstract concepts, and environments in which automated tests are designed and implemented.

Test Harness

A tool which enables the automation of tests. It refers to the system test drivers and other supporting tools that requires to execute tests. It provides stubs and drivers which are small programs that interact with the software under test.

Test Hierarchy

This is a class of terms describes the organization of tests into groups.

Test Methodology

This class of terms identifies the general methodology used by a test. Examples are White Box, Black Box.

Test result repository

Database of test results.

Test Results Trend-based

A matrix of correlation factors correlates test cases and code modules according to test results (verdict).

Test Roles

This class of terms identifies general roles and responsibilities for people relevant to testing.

Test Script

Automated test case. A single test script may be implemented with one or more test cases depending on the data.

Test Selection Method

This class of terms refers to the method used to select tests to be executed on a version of an EUT.

Test Session

Set of one or more test suites that are run together on a single build at a specific time.

Test Suite

Set of test cases that are run together on a single build at a specific time.

Test Trend

History of verdicts.

Test Type

The class which indicates the purpose of the test.

Test Version

The version of files used to test a specific build.

Tester

An individual who has the responsibility to test a system or service.

Testing Tools

Tools that verify code quality before passing the build.

The 4 Ts (Topology, Telemetry, Tracing, and Time)

Four pillars of Observability focusing on structure, signals, causality, and chronology.

The Advice Process

Any person deciding must seek advice from everyone meaningfully affected by the decision and people with expertise in the matter. Advice received must be taken into consideration, though it does not have to be accepted or followed. The objective of the advice process is not to form a consensus, but to inform the decision-maker so that they can make the best decision possible. Failure to follow the advice process undermines trust and unnecessarily introduces risk to the business.

The Checkbox Trap

The situation wherein an audit-centric perspective focuses exclusively on "checking the box" on compliance requirements without consideration for overall security objectives.

The Power of TED

The Power of TED* offers an alternative to the Karpman Drama Triangle with its roles of Victim, Persecutor, and Rescuer. The Empowerment Dynamic (TED) provides the antidote roles of Creator, Challenger, and Coach and a more positive approach to life's challenges.

The Sprint

A period of one month or less during which an increment of work is completed.

The Three Pillars of Empiricism

Three pillars uphold every implementation of empirical process control: transparency, inspection, and adaptation.

Theme

A theme provides a convenient way to indicate that a set of related epics have something in common, such as being in the same functional area. By assigning a financial value to Themes, managers can ensure the highest value is being delivered and that the project/program is aligned with its objectives and the strategic direction of the organization.

The Three Ways

Key principles of DevOps – Flow, Feedback, Continuous experimentation, and learning.

Theory of Constraints (ToC)

Methodology for identifying the most important limiting factor (that is, constraint) that stands in the way of achieving a goal and then systematically improving that constraint until it is no longer the limiting factor.

Thomas Kilmann Inventory (TKI)

Measures a person's behavioural choices under certain conflict situations.

Threat Agent

An actor, human or automated, that acts against a system with intent to harm or compromise that system. Sometimes also called a "Threat Actor".

Threat Detection

Refers to the ability to detect, report, and support the ability to respond to attacks. Intrusion detection systems and denial-ofservice systems allow for some level of threat detection and prevention.

Threat Intelligence

Information pertaining to the nature of a threat or the actions a threat may be known to be perpetrating. May also include "indicators of compromise" related to a given threat's actions, as well as a "course of action" describing how to remediate the given threat action.

Threat Modelling

A method that ranks and models potential threats so that the risk can be understood and mitigated in the context of the value of the application(s) to which they pertain.

Three Pillars of Observability

The three types of telemetry which include logs, traces and metrics.

Time-series data

A series of data points collected over time. Time-series data provide insight into how a system is behaving over time. Data that is collected over time and can be used to analyse trends and patterns in a system's behaviour.

Time to Insight Actioned

The time between having an idea, delivering it to the customer, learning and actioning the insight from that learning.

Time to Learning

The time between conceiving an idea and learning how it was received based on customer feedback.

Time to Market

The period of time between when an idea is conceived and when it is available to customers.

Time to Value

The measure of the time it takes for the business to realize value from a feature or service.

Time Tracking

Tools that allow for time to be tracked, either against individual issues or other work or project types.

Time Travel Topology

Topology map of an IT environment that keeps changing while the problem is happening. A change is often the cause of a problem so having time helps.

Timebox

The maximum duration of a Scrum event.

Toil

A kind of work tied to running a production service that tends to be manual, repetitive, automatable, tactical, devoid of enduring value.

Tool

This class describes tools that orchestrate, automate, simulate and monitor EUT's and infrastructures.

Toolchain

A philosophy that involves using an integrated set of complimentary task-specific tools to automate an end-to-end process (vs. a singlevendor solution).

Topology

A map that describes the set of relationships and dependencies between discrete components in an environment.

Touch Time

Touch Time is the actual time spent processing.

Traces

A collection of operations that represents a unique transaction handled by an application and its constituent services. The process of following a request or transaction through a system to understand how it is processed, in data that is visualized for a person. Tracing can be used to identify performance bottlenecks and diagnose issues.

Trace ID

A generated identifier (usually 32 hex characters) to identify a particular trace. Every span in the trace includes the same Trace ID.

Trace Propagation

The mechanism that moves trace context (Trace ID and Span ID) between services and processes. Between processes, it allows the downstream service to continue the trace.

Tracing

Tracing provides insight into the performance and health of a deployed application, tracking each function or microservice which handles a given request.

Traffic Volume

The amount of data sent and received by visitors to a service (for example, a website or API).

Training From the Back of the Room

An accelerated learning model in line with agile values and principles using the 4Cs instructional design “map” (Connection, Concept, Concrete Practice, Conclusion).

Transience

Transience in monitoring refers to the challenge of capturing and analysing short-lived events or states in dynamic systems.

Transformational Leadership

A leadership model in which leaders inspire and motivate followers to achieve higher performance by appealing to their values and sense of purpose, facilitating wide-scale organizational change. (State of DevOps Report, 2017)

Tribe Lead

A senior technical leader that has broad and deep technical expertise across all the squads’ technical areas. A group of squads working together on a common feature set, product, or service is a tribe in Spotify’s definitions.

Tribes

A collection of squads with a long-term mission that work on/in a related business capability.

Trojan (horses)

Malware that carries out malicious operations under the appearance of a desired operation such as playing an online game. A Trojan horse differs from a virus because the Trojan binds itself to non-executable files, such as image files, audio files whereas a virus requires an executable file to operate.

Trunk

The primary source code integration repository for a software product.

Unit Test

The purpose of the test is to verify code logic.

Unstructured Data

Unstructured data does not fit into any database structure, has no rules or format and is considered “free-form”, and it cannot be easily used by programs.

Unsupervised Learning

It refers to algorithms that learn patterns from unlabeled data. It works with unlabeled data and requires a lower degree of human intervention. There are three types to include clustering, association and dimensionality reduction.

Usability Test

The purpose of the test is to determine if humans have a satisfactory experience when using an EUT.

USE (Utilization, Errors, Duration) Metrics

A methodology for analyzing the performance of any system. It directs the construction of a checklist, which for server analysis can be used for quickly identifying resource bottlenecks or errors.

User

Consumer of IT services, or the identity asserted during authentication (also known as username).

User and Entity Behaviour Analytics (UEBA)

A machine learning technique to analyze normal and “abnormal” user behaviour with the aim of preventing the latter.

User Interface (UI)

The point of human-computer interaction and communication.

User Story

A brief statement used to describe a requirement from a user’s perspective. User stories are used to facilitate communication, planning, and negotiation activities between the stakeholders and the Agile Service Management Team.

Utilization, Saturation, and Errors (USE) metrics

Three resource metrics used to analyse system performance.

Value

One of the 5Vs of Big Data. Value refers to the ability to extract meaningful insights and value from Big Data.

Value Added Time

The amount of time spent on an activity that creates value (for example, development, testing).

Value Cycle

The lifecycle stages of the value stream from ideation to value realization.

Value Efficiency

Being able to produce value with the minimum amount of time and resources.

Value Stream

All of the activities needed to go from a customer or end-user request to a delivered product or service.

Value Stream Intelligence (VSI)

Learning how to connect and leverage customer, business, and organizational data to gain systemic, actionable insights for continuous improvement.

Value Stream Map

Visually depicts the end-to-end flow of activities from the initial request to value creation for the customer.

Value Stream Mapping

A lean tool that depicts the flow of information, materials, and work across functional silos with an emphasis on quantifying waste, including time and quality.

Value Stream Management

Value Stream Management is a combination of people, processes, and technology that maps, optimizes, visualizes, measures, and governs business value flow through heterogeneous software delivery pipelines from idea through development and into production.

Value Stream Management Platform

Software that manages value streams.

Value Stream Network

The entire interconnected ecosystem of all value streams and their supporting systems within an organization, often across multiple products and departments.

Value Stream Sponsor

A value stream sponsor is an executive or senior leader who owns the ultimate accountability for the business outcomes, funding, and strategic direction of a specific value stream.

Variable Speed IT

An approach where traditional and digital processes co-exist within an organization while moving at their own speed.

Variety

One of the 5Vs of Big Data. Variety refers to the diverse types and formats of data that are encountered in Big Data environments.

Velocity

One of the 5Vs of Big Data. Velocity refers to the speed at which data is generated, processed, and analysed.

Veracity

One of the 5Vs of Big Data. Veracity refers to the quality, accuracy, and reliability of data.

Verdict

Test result classified as Fail, Pass, or Inconclusive.

Version control tools

Ensure a 'single source of truth' and enable change control and tracking for all production artifacts.

Vertical Scaling

Computing resources are scaled higher to increase processing speed, for example, using faster computers to run more tasks faster.

Virtual Private Cloud (VPC)

A logically isolated cloud environment within a public cloud provider.

Virtual Private Network (VPN)

An encrypted connection that secures traffic between devices and private networks over public internet.

Virus (Computer)

Malicious executable code attached to a file that spreads when an infected file is passed from system to system that could be harmless (but annoying), or it could modify or delete data.

Visualization

The process of presenting data in a visual format to help engineers understand and analyze a system's behaviour. Visualization can include graphs, charts, and dashboards.

Voice of the Customer (VOC)

A process that captures and analyzes customer requirements and feedback to understand what the customer wants.

Volume

One of the 5Vs of Big Data. Volume refers to the vast amount of data that is generated and collected.

Vulnerability

A weakness in a design, system, or application that can be exploited by an attacker.

Vulnerability Intelligence

Information describing a known vulnerability, including affected software by version, the relative severity of the vulnerability (for example, does it result in an escalation of privileges for a user role, or does it cause a denial of service), the exploitability of the vulnerability (how easy/ hard it is to exploit), and sometimes current rate of exploitation in the wild (is it being actively exploited or is it just theoretical). This information will also often include guidance on what software versions are known to have remediated the described vulnerability.

Vulnerability management

The process of identifying and remediating vulnerabilities.

Wait Time

The amount of time wasted on waiting for work (e.g., waiting for development and test infrastructure, waiting for resources, waiting for management approval).

Waste (Lean)

Any activity that does not add value to a process, product or service.

Water-scrum-fall

A hybrid approach to application lifecycle management that combines waterfall and Scrum development can complete in a given amount of time.

Waterfall (Project Management)

A linear and sequential approach to managing software design and development projects in which progress is seen as flowing steadily (and sequentially) downwards (like a waterfall).

Weakness

An error in software that can be exploited by an attacker to compromise the application, system, or the data contained therein. Also called a vulnerability.

Web Application Firewall (WAF)

Tools that examine traffic being sent to an application and can block anything that looks malicious.

Web IDE

Tools that have a web client integrated development environment. Enables developer productivity without having to use a local development tool.

Westrum (Organization Types)

Ron Westrum developed a typology of organizational cultures that includes three types of organizations: Pathological (poweroriented), Bureaucratic (rule-oriented) and Generative (performance-oriented).

White-Box Testing (or Clear-, Glass-, Transparent-Box Testing or Structural Testing)

Test cases use extensive knowledge of the internal design structure or workings of an application, as opposed to its functionality (that is, Black-Box Testing).

Whitelisting

Application whitelisting is the practice of specifying an index of approved software applications that are permitted to be present and active on a computer system.

Wicked Questions

Wicked questions are used to expose the assumptions which shape our actions and choices. They are questions that articulate the embedded, and often contradictory assumptions, we hold about an issue, a problem or a context.

Wiki

Knowledge sharing can be enabled by using tools like Confluence which create a rich Wiki of content

Wilber's Quadrants

A model that recognises four modes of general approach for human beings. Two axes are used: on one axis people tend towards individuality OR collectivity.

Work in Progress (WIP)

Any work that has been started but has not been completed.

Workaround

A temporary way to reduce or eliminate the impact of incidents or problems. May be logged as a known error in the Known Error Database. (ITIL definition)

World Café

Is a structured conversational process for knowledge sharing in which groups of people discuss a topic at several tables, with individuals switching tables periodically and getting introduced to the previous discussion at their new table by a "table host".

Worms (Computer)

Worms replicate themselves on a system by attaching themselves to different files and looking for pathways between computers. They usually slow down networks and can run by themselves (where viruses need a host program to run).

W3C Standard

W3C standards define an Open Web Platform for application development that has the unprecedented potential to enable developers to build rich interactive experiences, powered by vast data stores, that are available on any device.

Zipkin

An open-source distributed tracing system for monitoring microservice interactions.

Notes

Notes

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THE LANGUAGE OF COLLABORATION

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